北京开源芯片研究院
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香山开源处理器社区
XiangShan Open-Source Processor Community

Introduction to the XS-Gem5 Simulator

The XiangShan Team

MICRO 25 @ Seoul, Korea

October 19, 2025

What we will cover in this tutorial

- **XS-Gem5**
  - Introduction:
  - Background: Why build such a simulator?
  - Calibration & Architecture: How is the simulator designed?
  - Development Tools: What tools are we using at the current stage to improve development efficiency?
  - Case Study: A concrete example demonstrating how the simulator is used.

What we will cover in this tutorial

- **XS-Gem5**
  - **Introduction:**
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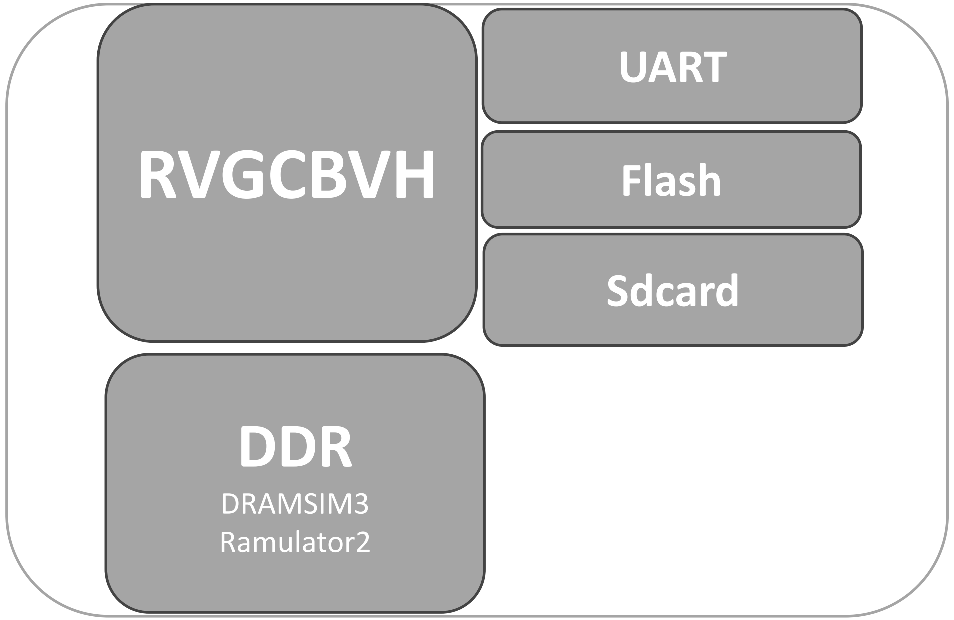
Introduction

- **XS-Gem5 Architectural Simulator**
  - Built on the open-source GEM5 simulator
  - Simulation speed: ~100 KIPS
  - Enables rapid design space exploration

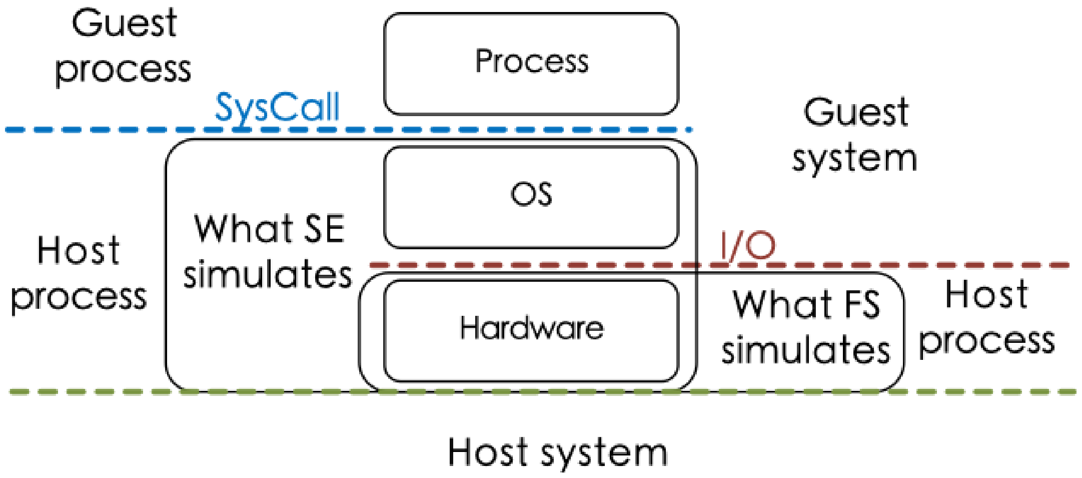
Introduction

- **Features**
  - Se simulation
  - Full-system simulation
    - RVGCpt: XiangShan's RISC-V generic checkpoint
    - Online Difftest
- Estimated SPECCPU **06 &&17** overall score with SimPoint
- **Frontend**
  - micro-architecture calibration
- **Backend/Cache**
  - latency/throughput calibration
  - Pipeline && Issue queue && Regfile
- **Memory system**
  - Prefetchers
  - Parallel PTW / L2 TLB / TLB prefetching
  - Memory bandwidth and memory controller calibration
  - 1MB L2
  - 16MB L3
  - DDR4 3200 dual-channel **OR** DDR5 3200 AN



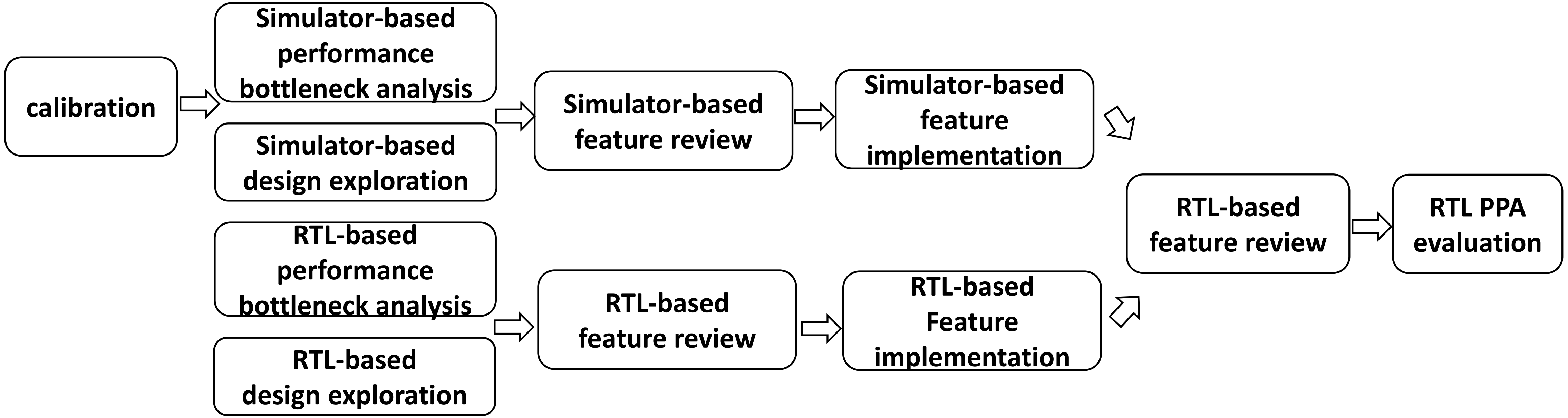
Support Full-system && Se simulation

- **Se mode**
  - Support part microbenchmark
- **Full-system**
  - Avoid hacky system calls in SE (system call emulation)
  - **Accurate system call modeling**
    - Performance of serializing instructions
  - **Accurate virtual memory behavior**
    - Random VA → PA mapping
    - Accurate TLB/PTW modelling
    - Recommend using fs.



Performance Exploration Workflow on an Architectural Simulator

- The results obtained from architectural simulator-based exploration should provide **guidance** for performance iteration on XiangShan.
- Findings derived from a calibrated simulator are more likely to be validated on RTL.
- Performance iteration workflow guided by the simulator:



Schedule

- **XS-Gem5**
  - Introduction :
  - **Background: Why build such a simulator?**
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Background - Rapid performance iteration

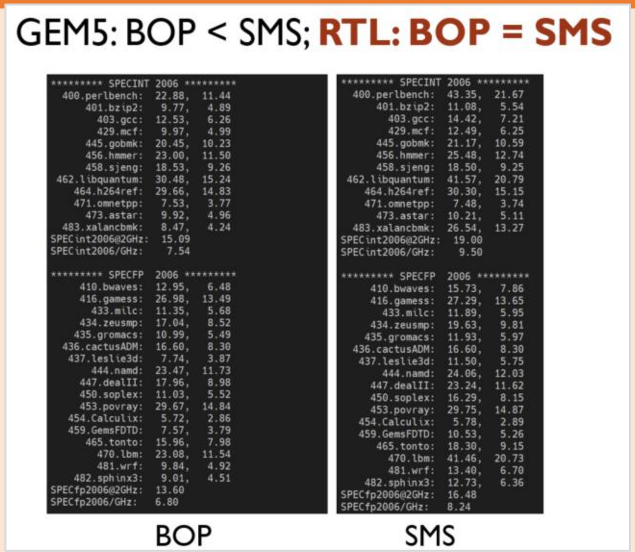
- Three generations of architectural upgrades within three years:
  - Yanqihu → Nanhu → Kunminghu
  - 7 /GHz → 10 /GHz → 15 /GHz

- High development efficiency is achieved by architectural simulation co-design
  - Flexible, easy to modify
  - Fast simulation time
  - Enable evaluation of new features prior to RTL implementation
  - Enable evaluation of optimal configurations in vast design space

Background - Pitfalls in simulation methodologies

"While absolute accuracy for simulator may be low, relative performance trend can be accurate"
The assumption that "the overall trend remains reliable" does not always hold true [1]

Problem: The "trend myth" of simulators


GEM5: BOP < SMS; **RTL: BOP = SMS**

```
********* SPECINT 2006 *********
  400.perlbench:  22.88,  11.44
      401.bzip2:   9.77,   4.89
        403.gcc:  12.53,   6.26
        429.mcf:   9.97,   4.99
     445.gobmk:  20.45,  10.23
     456.hmmer:  23.00,  11.50
     458.sjeng:  18.53,   9.26
 462.libquantum:  30.48,  15.24
   464.h264ref:  29.66,  14.83
   471.omnetpp:   7.53,   3.77
     473.astar:   9.92,   4.96
  483.xalancbmk:   8.47,   4.24
SPECint2006@2GHz:  15.09
SPECint2006/GHz:    7.54

********* SPECFP  2006 *********
    410.bwaves:  12.95,   6.48
    416.gamess:  26.98,  13.49
      433.milc:  11.35,   5.68
    434.zeusmp:  17.04,   8.52
   435.gromacs:  10.99,   5.49
  436.cactusADM:  16.60,   8.30
   437.leslie3d:   7.74,   3.87
      444.namd:  23.47,  11.73
    447.dealII:  17.96,   8.98
    450.soplex:  11.03,   5.52
    453.povray:  29.67,  14.84
   454.Calculix:   5.72,   2.86
   459.GemsFDTD:   7.57,   3.79
     465.tonto:  15.96,   7.98
       470.lbm:  23.08,  11.54
       481.wrf:   9.84,   4.92
    482.sphinx3:   9.01,   4.51
SPECfp2006@2GHz:  13.60
SPECfp2006/GHz:   6.80
```

BOP

```
********* SPECINT 2006 *********
  400.perlbench:  43.35,  21.67
      401.bzip2:  11.08,   5.54
        403.gcc:  14.42,   7.21
        429.mcf:  12.49,   6.25
     445.gobmk:  21.17,  10.59
     456.hmmer:  25.48,  12.74
     458.sjeng:  18.50,   9.25
 462.libquantum:  41.57,  20.79
   464.h264ref:  30.30,  15.15
   471.omnetpp:   7.48,   3.74
     473.astar:  10.21,   5.11
  483.xalancbmk:  26.54,  13.27
SPECint2006@2GHz:  19.00
SPECint2006/GHz:    9.50

********* SPECFP  2006 *********
    410.bwaves:  15.73,   7.86
    416.gamess:  27.29,  13.65
      433.milc:  11.89,   5.95
    434.zeusmp:  19.63,   9.81
   435.gromacs:  11.93,   5.97
  436.cactusADM:  16.60,   8.30
   437.leslie3d:  11.50,   5.75
      444.namd:  24.06,  12.03
    447.dealII:  23.24,  11.62
    450.soplex:  16.29,   8.15
    453.povray:  29.75,  14.87
   454.Calculix:   5.78,   2.89
   459.GemsFDTD:  10.53,   5.26
     465.tonto:  18.30,   9.15
       470.lbm:  41.46,  20.73
       481.wrf:  13.40,   6.70
    482.sphinx3:  12.73,   6.36
SPECfp2006@2GHz:  16.48
SPECfp2006/GHz:   8.24
```

SMS

- **High-performance processors have complex microarchitectural details.**
- **Uncalibrated simulators fail to reflect the correct design trends.**

- Uncalibrated simulators → **fail to reflect performance characteristics**:
  - Misleading performance insights
  - Wasted engineering efforts
  - Low return-on-cost design

Calibration is crucial for the simulator.

[1]Tony Nowatzki, Jaikrishnan Menon, Chen-Han Ho, and Karthikeyan Sankar-alingam. 2015. Architectural simulators considered harmful. IEEE Micro 35, 6(2015), 4–12.

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Microbenchmarks for micro-arch calibration

- **Principle:** Calibrate the architectural model by extracting key microarchitectural parameters of the processor through external observation.
- **Work:** Designed and implemented a set of microbenchmarks for simulator RTL calibration.

Pipeline

- Instruction operation latency
- Operand-dependent division operations, scheduling latency
- Instruction scheduling latency
- Decode bandwidth
- Rename bandwidth
- Dispatch bandwidth

Cache

- Load-to-use latency (L1/L2/L3 hit)
- Load-Load schedule latency (L1/L2/L3hit)
- L1/L2/L3 access latency
- L1/L2/L3 access bandwidth
- Stream store access pattern

Virtual

- L2 TLB access latency
- L2 TLB access latency to cache and memory at all levels
- Stride Prefetch
- Stream Prefetch

Branch prediction

- Penalty of flushing instructions after branch misprediction
- Long instruction sequence test without jumps

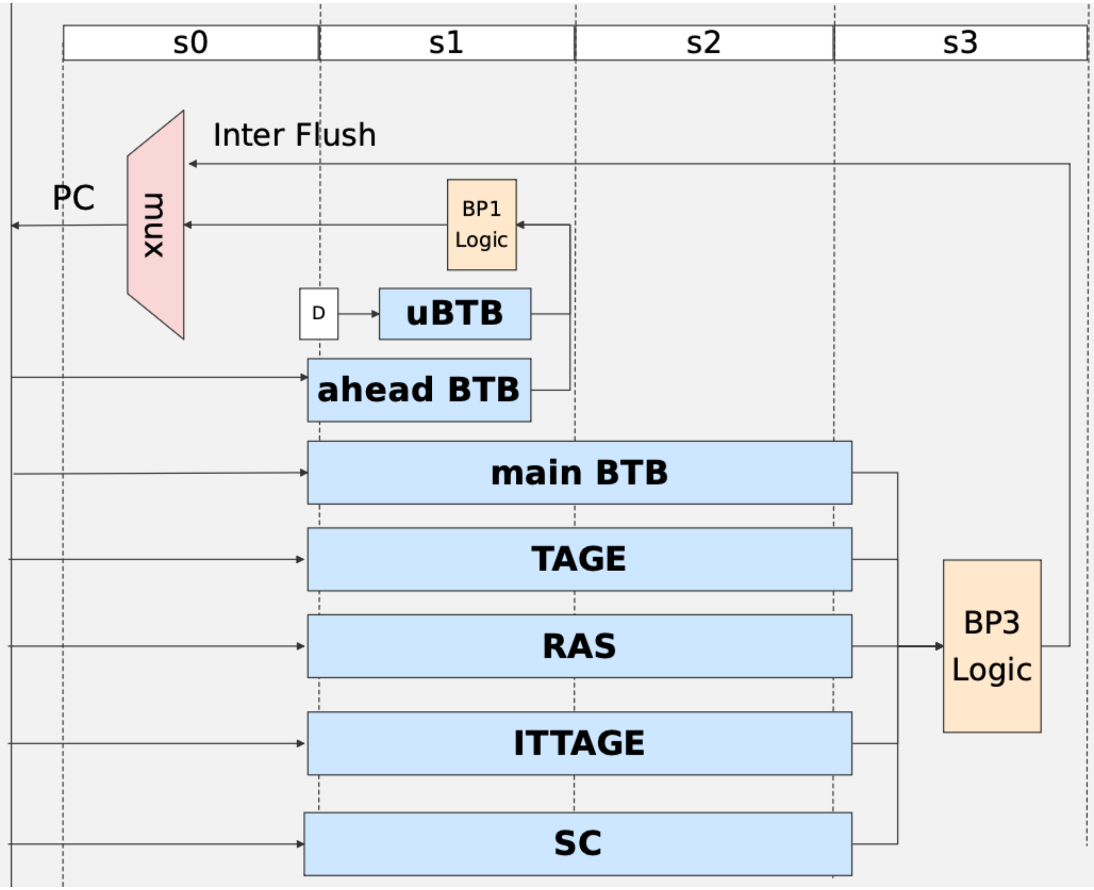
Uncalibrated points

- Dispatch logic is too ideal in architecture simulator
- Division latency and operand relationship is wrong in architecture simulator
- RTL CSR_MINSTRET is wrong
- The store-load delay of architecture simulator and RTL is inconsistent
- Architecture simulator does not implement conflict detection logic
- Architecture simulator and RTL memory access latency are inconsistent
- Architecture simulator prefetch parameters are wrong

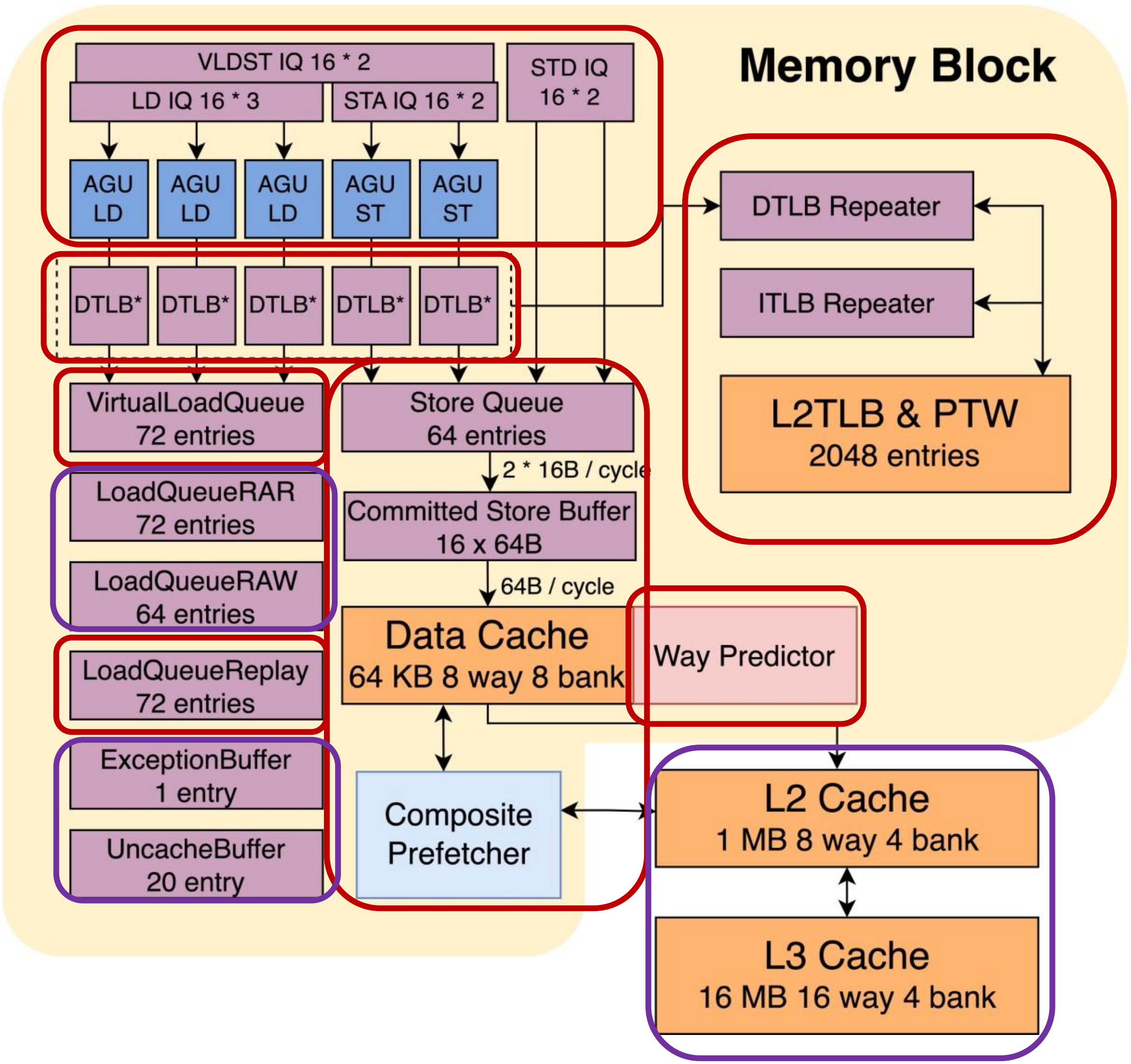
Infrastructure: the entire process of automated test running, and reporting results

Frontend

- **Decoupled Frontend**
  - Hides latency and boosts pipeline efficiency
- **Advanced Predictor Suite**
  - Integrates TAGE, SC, and others
  - TAGE
    - Expanded from 4 to 10 tables
    - History up to 500 bits
  - SC
    - Six correction tables
- **Highly scalable**
  - Supports multiple decoupled branch predictors
  - Configurable latency



Memblock



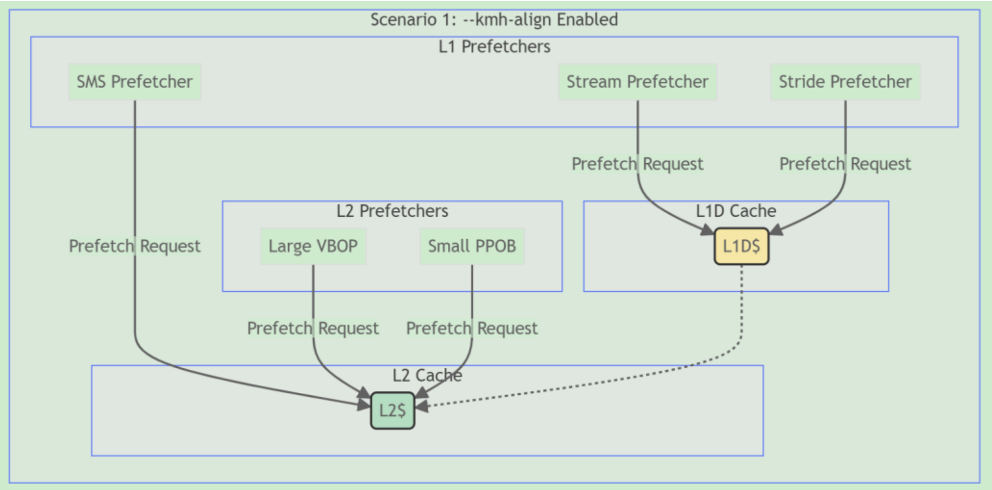
- What we have calibrated
  - Load & Store pipeline details[1]
  - Store STA/STD split[1]
  - Most Load replay scenarios modeling[1]
  - Prefetcher [4] & Way Predictor [5]
  - StoreBuffer [1]
  - MMU [2,3]

- Differences still
  - No detailed LoadQueue split
  - L2 & L3 Cache hardware logic details

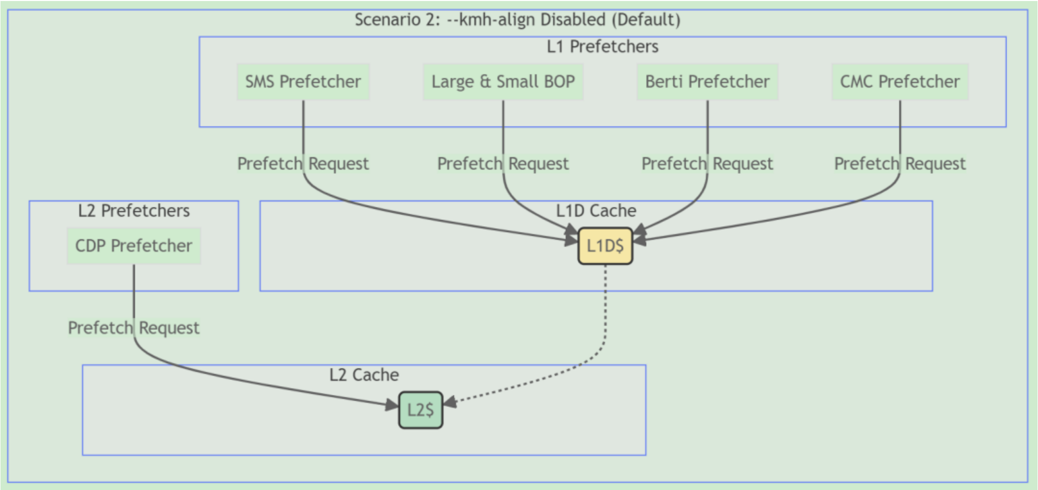
Code details
*[1] src/cpu/o3/lsq**
*[2] src/arch/riscv/tlb**
*[3] src/arch/riscv/pagetable_walker**
*[4] src/mem/cache/prefetch/**
*[5] src/mem/cache/way_prediction_policies/**

Prefetcher

- Calibrated configuration with RTL (vias --kmh-align)
  - L1 Prefetchers: Stream + Stride
  - L2 Prefetchers: SMS + Large Virtual BOP + Small Physical BOP
- SOTA prefetch algorithms & mechanisms
  - IPCP, SPP, Berti, CDP, and temporal prefetcher
  - Prefetching request **offloading**: passive & active
  - Dynamic Prefetching control



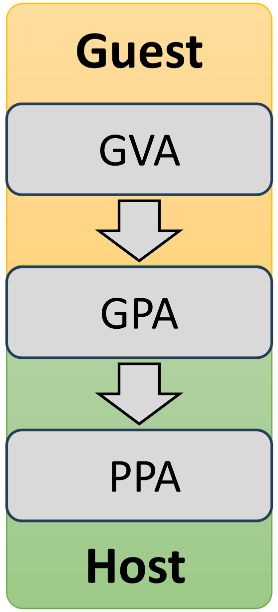
Well Calibrated



Better Performance

Support Hypervisor Extension

- **Functionality**
- **Architecture**
  - Supports SV39 and SV39x4
  - Supports SV48 and SV48x4
  - Supports two-stage address translation
  - Supports H-extension exception handling flow
  - Supports execution of H-extension CSR instructions
  - Supports interrupt controller Infra
- **Architectural calibration**
  - L1 TLB: 48-entry ITLB and DTLB
  - L2 TLB: L0, L1, and L2 store corresponding levels of the page table
- **Difference**
  - PTW (Page Table Walker): Idealized PTW model with no restriction on parallelism



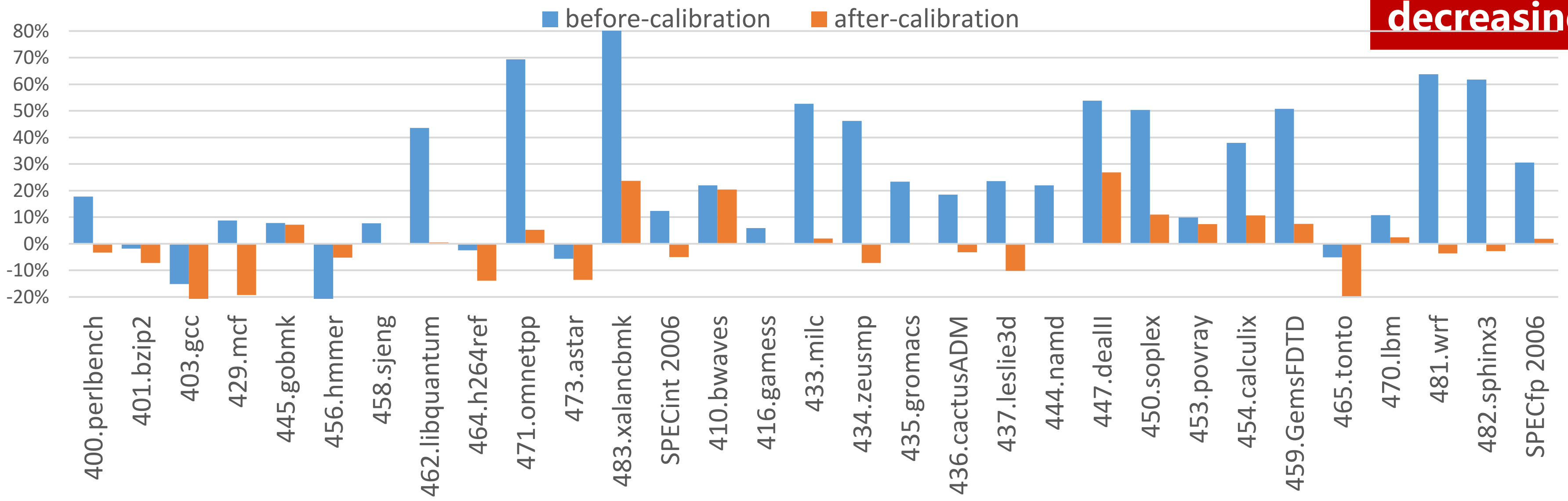
Support for multiple DDR models

- Integration of **DRAMSim3**
  - Provides mainstream open-source DRAM modeling capability
  - Enables basic latency and bandwidth modeling for DDR3/DDR4
- Integration of **Ramulator2**
  - A new generation of flexible and extensible DRAM simulator
  - Supports multiple scheduling and row policies (e.g., FRFCFS, OpenRowPolicy)
  - Capable of modeling DDR3/DDR4/DDR5
- **Flexible configuration mechanism**
  - Switch between DRAMSim3 and Ramulator2 with a simple toggle
  - One-click configuration to switch between DDR4 and DDR5 models
  - Supports diverse simulation scenarios to meet different requirements

Calibration

- Performance deviation between calibrated GEM5 and RTL on SPEC06:
  - SPEC06 Integer: 1.79% deviation between GEM5 and RTL
  - SPEC06 Floating Point: 2.69% deviation between GEM5 and RTL

The gap in performance difference is gradually decreasing



Current SPECCPU 2006 Score

- Estimated with RVGCpt + SimPoint
- Memory simulated with DRAMSim3:
  - DDR4 3200 Dual-channel
- Cache:
- 64kB Dcache+1 MB L2 +16MB L3
- Prefetcher:
  - **Stream + Stride + SMS + BOP**
  - With IPCP/SPP/Temporal **off**
- **SPEC int 2006 score: 15.40/GHz**
- **SPEC fp 2006 score: 15.04/GHz**
  - Compilation: **GCC 12 o3 base**

perlbench	12.619
bzip2	8.89
gcc	15.855
mcf	21.418
gobmk	10.594
hmmer	14.178
sjeng	10.711
libquantum	40.5
h264ref	19.019
omnetpp	14.516
astar	10.207
xalancbmk	25.439
Int per GHz	15.40004723

bwaves	22.405
gamess	13.879
milc	16.25
zeusmp	15.056
gromacs	11.663
cactusADM	14.761
leslie3d	14.864
namd	9.508
dealII	25.328
soplex	20.564
povray	18.502
calculix	5.575
GemsFDTD	16.887
tonto	12.005
lbm	28.414
wrf	13.383
sphinx3	13.497
FP per GHz	15.04328747

Architecture exploration case: SMS algorithm optimization

- **Propose several SMS prefetch optimization algorithms**
  - Timeliness: PHT prefetch update algorithm guided by access timing
  - Coverage: PHT prefetch confidence calculation method guided by access frequency
- **Choose the features with better effect on the simulator and complete the RTL implementation**
- **The total SPEC06 score increased by 0.35% on the simulator and 0.4% on RTL**

	baseline	sms improve	
SPECINT			
400.perlbench:	13.043	13.085	0.32%
401.bzip2:	8.808	8.81	0.02%
403.gcc:	17.436	17.484	0.27%
429.mcf:	20.979	21.072	0.44%
445.gobmk:	9.92	9.921	0.01%
456.hmmer:	13.594	13.594	0.00%
458.sjeng:	10.364	10.334	-0.29%
462.libquantum:	42.172	42.092	-0.19%
464.h264ref:	20.327	20.337	0.05%
471.omnetpp:	12.793	13.004	1.62%
473.astar:	19.124	19.066	-0.30%
483.xalancbmk:	24.653	24.672	0.08%
	16.101485	16.12901676	0.17%
SPECFP			
410.bwaves:	25.078	25.723	2.51%
416.gamess:	14.693	14.698	0.03%
433.milc:	12.871	13.297	3.20%
434.zeusmp:	18.688	18.497	-1.03%
435.gromacs:	12.796	12.869	0.57%
436.cactusADM:	16.398	16.51	0.68%
437.leslie3d:	15.441	15.834	2.48%
444.namd:	11.554	11.569	0.13%
447.dealII:	19.355	19.399	0.23%
450.soplex:	16.908	16.848	-0.36%
453.povray:	18.049	18.093	0.24%
454.Calculix:	6.205	6.204	-0.02%
459.GemsFDTD:	11.336	11.36	0.21%
465.tonto:	11.689	11.679	-0.09%
470.lbm:	33.972	33.938	-0.10%
481.wrf:	11.712	11.799	0.74%
482.sphinx3:	18.279	18.294	0.08%
	15.15381473	15.24000599	0.57%

Simulator-based performance bottleneck analysis → Simulator-based feature implementation → Simulator-based feature review → RTL-based feature implementation → RTL-based feature review → Merge to master

Processor architecture exploration guided by simulator end to end

Schedule

- **XS-Gem5**
  - Overall Structure:
  - Background: Why build such a simulator?
  - Motivation: Why is this simulator necessary?
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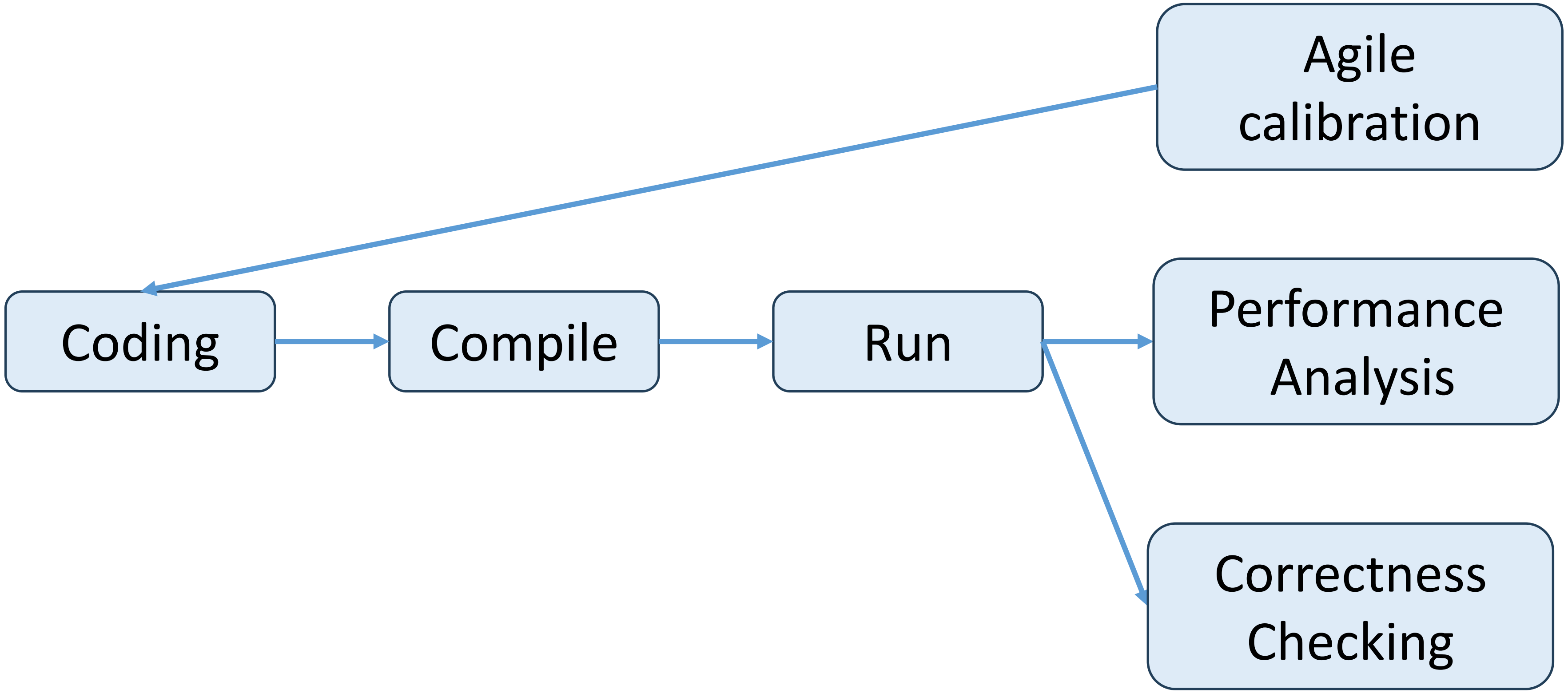
Introduction to Development Tools

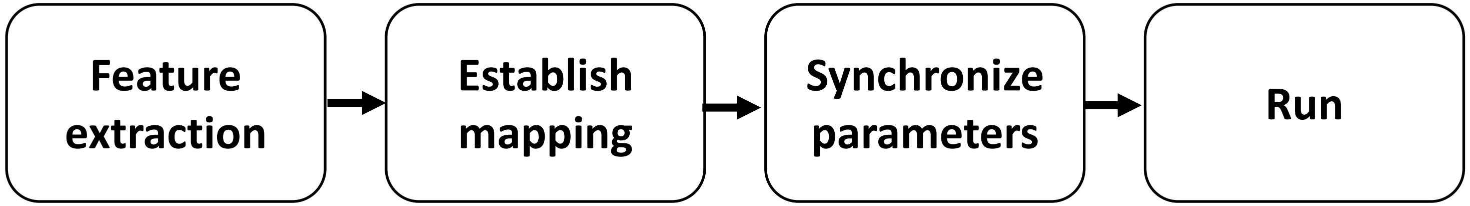
- Development Workflow on an Architectural Simulator



- **Auto-P-calibrate**
- **PerfCCT**
- **Top-Down**
- **DiffTest**

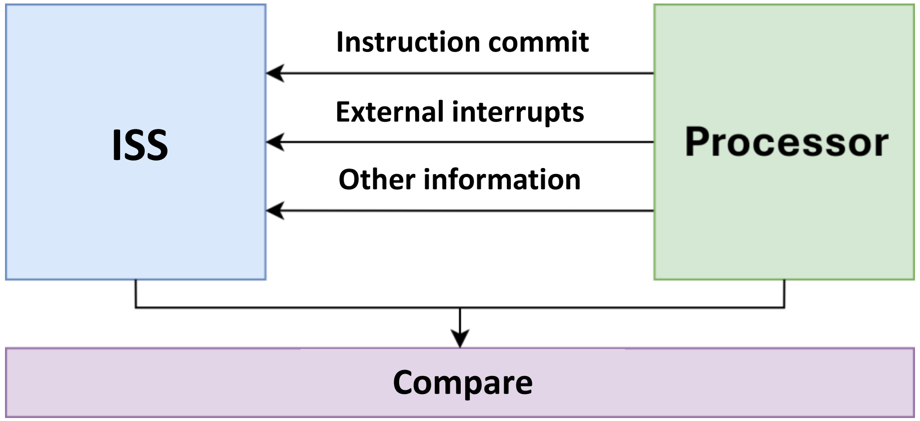
Automated parameter calibration

- RTL has many parameters, enabling dynamic calibration during development
- Automated scripts extract key architectural parameters from RTL
- Establish XS-RTL - XS-GEM5 parameter mapping
- Synchronize parameters to XS-GEM5 automatically
- Basic functionality implemented, supporting dynamic parameter calibration



DiffTest: Efficient Debugging

- **Accelerating Architectural Simulator Debugging**
  - Implemented the DIFFTEST framework
  - Online ISA-level behavior verification
  - Uses NEMU/Spike as the reference model (REF)
  - Compares register states between the architectural simulator and REF → triggers error or continues execution
  - Quickly captures the context of instruction-level mismatches



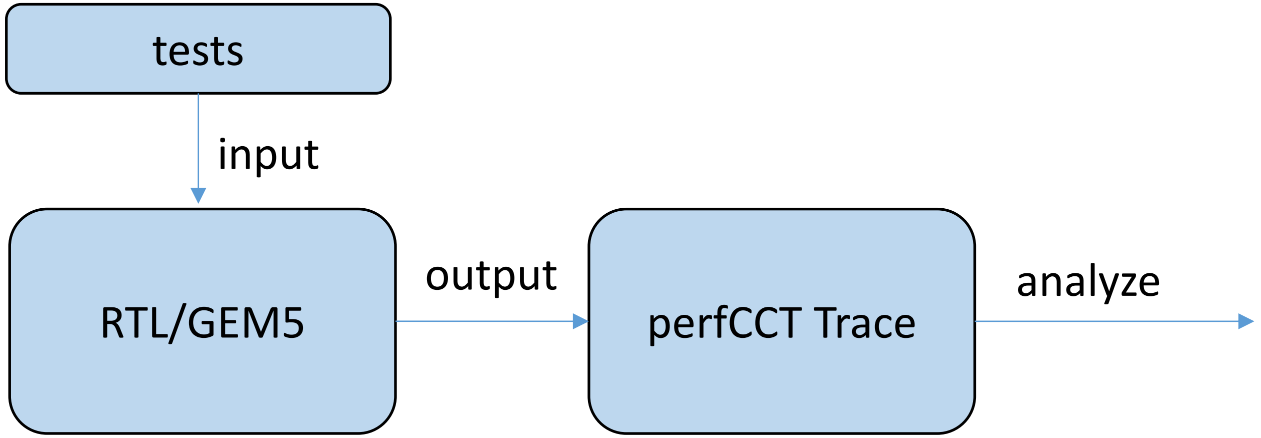
Basic architecture

```
while (1) {
    icnt = gem5_step();
    ref_step(icnt);
    r1s = gem5_getregs();
    r2s =  ref_getregs();
    if (r1s != r2s) { abort(); }
}
```

Online Comparison Mechanism

PerfCCT: Trace analysis tool

- **Track the life cycle of an instruction**
- **Life cycle:**
  - fetch/decode/rename/dispatch/issue/arbitration/execute/writeback/commit
- **Extract Hot loops from Traces after running which output:**
  - How many times each basic block appears
  - The proportion of each basic block in the total execution cycle
  - The Instructions of the basic block

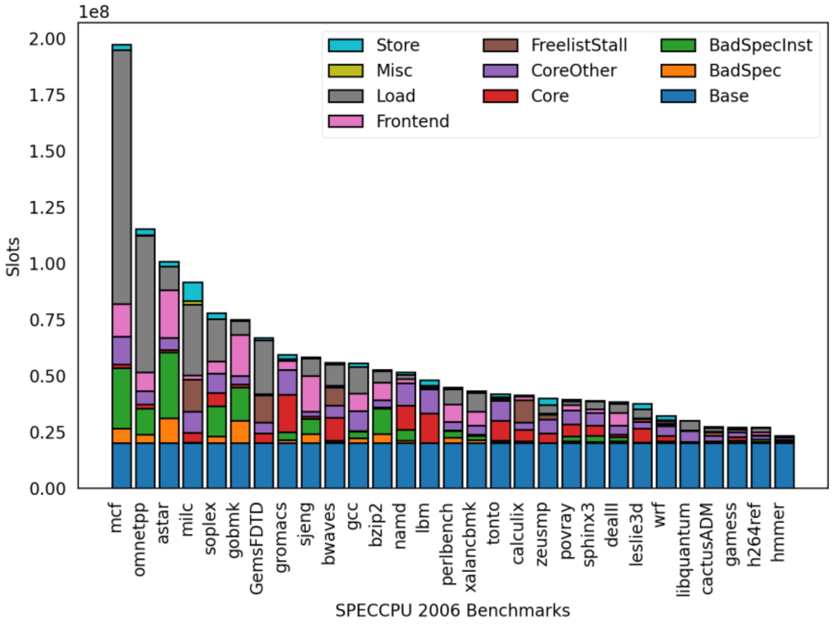


It is convenient for us to understand the characteristics of the test

```
Top 10 most common basic blocks:
Total cycle: 3040633
Count: 65265 (42.37%) cycle: 2182167 ratio (71.77%)
Instructions:
  0xaae98: add s1, a7, t0
  0xaae9c: fld fa4, -8(t4)
  0xaaea0: fld ft10, -32(a7)
  0xaaea4: fld fa1, -104(s1)
  0xaaea8: fld fa6, -96(t1)
  0xaaeac: fmul_d ft10, fa4, ft10
  0xaaeb0: fmul_d fa1, fa4, fa1
  0xaaeb4: fld ft4, -24(a1)
  0xaaeb8: fmul_d fa6, fa4, fa6
  0xaaebc: fld ft0, 48(a1)
  0xaaec0: fmul_d ft4, fa4, ft4
  0xaaec4: c_fld fa0, 120(a1)
  0xaaec6: fmul_d ft10, ft10, fs5
  0xaaeca: fmul_d fa1, fa1, fs11
  0xaaece: fmul_d ft0, fa4, ft0
  0xaaed2: fmul_d fa6, fa6, fs8
  0xaaed6: fmul_d fa0, fa4, fa0
  0xaaeda: fmul_d ft4, ft4, fs8
  0xaaede: fld fs0, 40(a7)
  0xaaee2: fmul_d ft10, ft10, fs11
  0xaaee6: fadd_d fa1, fa1, fa5
  0xaaeea: fmul_d ft0, ft0, fs8
  0xaaeee: fmul_d fa6, fa6, fs11
  0xaaef2: fmul_d fa0, fa0, fs8
  0xaaef6: fmul_d ft4, ft4, fs5
  0xaaefa: fmul_d fs10, fa4, fs0
  0xaaefe: fld fa5, -88(t1)
  0xaaf02: fadd_d fa1, fa1, ft10
  0xaaf06: fmul_d ft0, ft0, fs3
```

Top-Down: quickly identify performance bottlenecks

- Hierarchical Performance Counters
- Attribute instruction stalls to architectural components
- Top-down decomposition of processor performance events
- Quickly and accurately identify performance bottlenecks
- **Adapt Intel's Top-Down methodology to XiangShan design**
  - Optimize for RISC-V ISA
  - Enhance counters for XiangShan microarchitecture
  - Refine hierarchical Top-Down model design



Performance Bottleneck Statistics Visualization Example
Identify Performance Bottlenecks at the Macro Level

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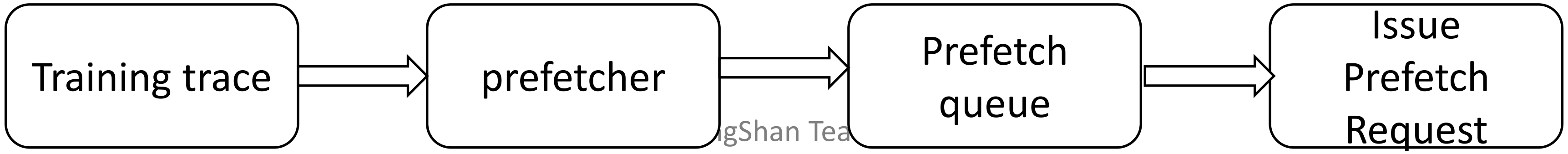
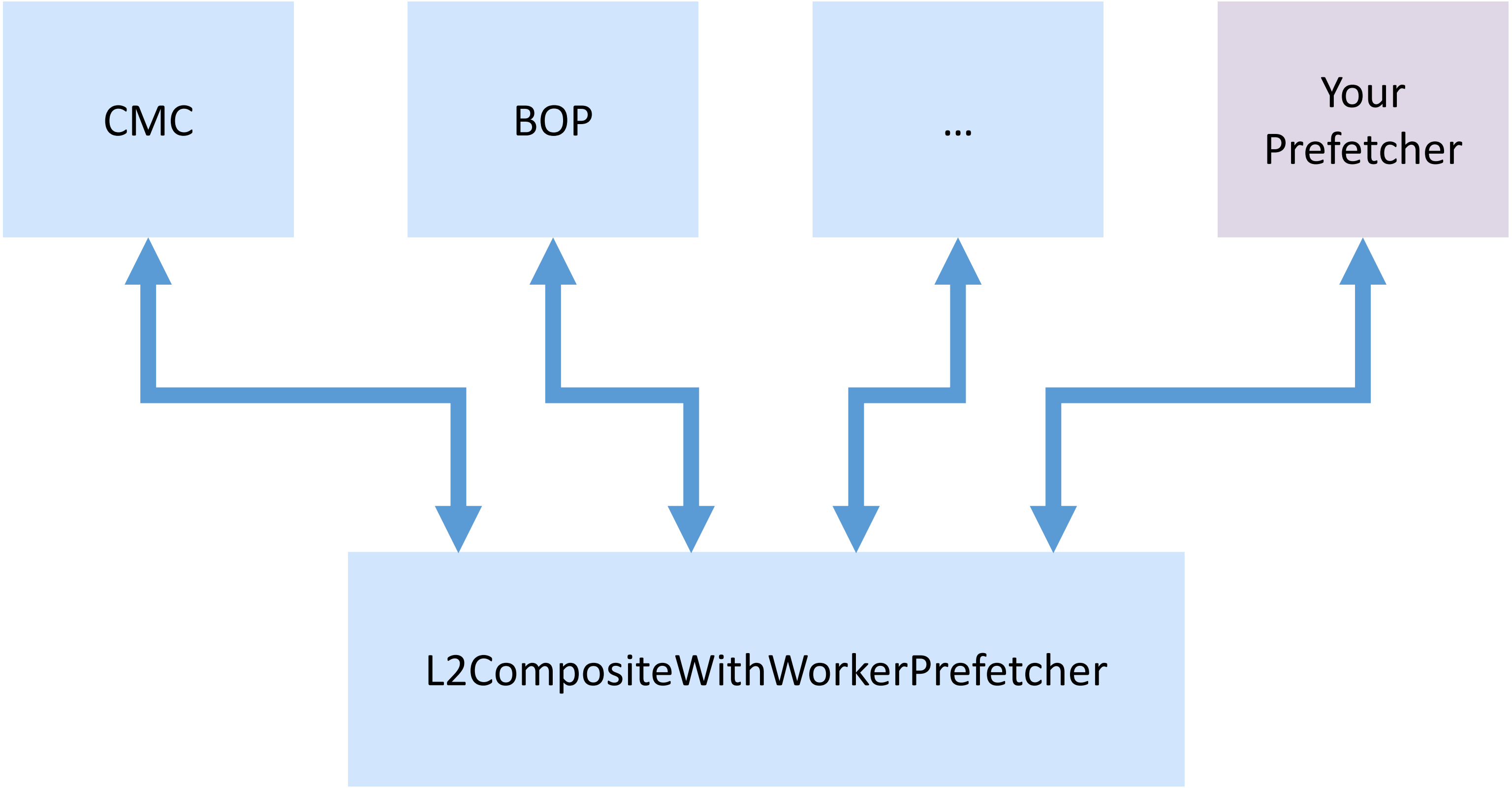
Case Study: How to add a new feature

- Example: Integrating a new prefetching algorithm for evaluation
- In XS-GEM5, prefetch execution is organized into several components. Adding a new prefetcher involves the following steps:
  - **Definition**
    - Define the new prefetcher in prefetcher.py.
  - **Configuration Switch**
    - Set whether the prefetcher is enabled by default in the configuration system.
  - **Declaration**
    - Declare the prefetcher within the simulator’s prefetching framework.
  - **Implementation**
    - Implement the logic of the prefetching algorithm.
  - **Invocation**
    - Integrate the prefetcher into the execution flow by registering the call logic within the existing prefetching framework.



Case Study: How to add a new feature



Case Study: How to add a new feature

- 1. Define the prefetcher in prefetcher.py
  - **Prefetcher Specification:**
    - Specify key parameters such as prefetch distance, trigger conditions.
  - **Replacement Policy:**
    - Implement internal management strategies such as entry replacement or priority handling within the prefetcher.

```
diff --git a/src/mem/cache/prefetch/Prefetcher.py b/src/mem/cache/prefetch/Prefetcher.py
index d2cce7eb4d..a566c2a648 100644
--- a/src/mem/cache/prefetch/Prefetcher.py
+++ b/src/mem/cache/prefetch/Prefetcher.py
@@ -929,6 +929,57 @@ class CMCPrefetcher(QueuedPrefetcher):
             "Enable prefetch database"
     )

+
+class DespacitoStreamPrefetcher(QueuedPrefetcher):
+    type = "DespacitoStreamPrefetcher"
+    cxx_class = "gem5::prefetch::DespacitoStreamPrefetcher"
+    cxx_header = "mem/cache/prefetch/despacito_stream.hh"
+
+    use_virtual_addresses = False
class XSCompositePrefetcher(QueuedPrefetcher):
    type = "XSCompositePrefetcher"
    cxx_class = 'gem5::prefetch::XSCompositePrefetcher'
@@ -1108,9 +1159,12 @@ class L2CompositeWithWorkerPrefetcher(CompositeWithWorkerPrefetcher):
                        "Large BOP used in composite prefetcher ")
    bop_small = Param.BOPPrefetcher(SmallBOPPrefetcher(is_sub_prefetcher=True),
                        "Small BOP used in composite prefetcher ")
+   despacito_stream = Param.DespacitoStreamPrefetcher(DespacitoStreamPrefetcher(is_sub_prefetcher=True),
+                       "DespacitoStream used in composite prefetcher")
    enable_bop = Param.Bool(False, "Enable BOP")
    enable_cdp = Param.Bool(True, "Enable CDP")
    enable_cmc = Param.Bool(False, "Enable CMC")
+   enable_despacito_stream = Param.Bool(True, "Enable despacito stream")

class L3CompositeWithWorkerPrefetcher(CompositeWithWorkerPrefetcher):
    type = 'L3CompositeWithWorkerPrefetcher'
```

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- 2. Enable or disable the prefetcher in prefetcherConfig.py

```
diff --git a/configs/common/PrefetcherConfig.py b/configs/common/PrefetcherConfig.py
index 9467763251..20d873d3da 100644
--- a/configs/common/PrefetcherConfig.py
+++ b/configs/common/PrefetcherConfig.py
@@ -65,6 +65,7 @@ def create_prefetcher(cpu, cache_level, options):
             prefetcher.enable_cmc = False
             prefetcher.enable_bop = True
             prefetcher.enable_cdp = False
+            prefetcher.enable_despacito_stream = False
             prefetcher.bop_large = XSVirtualLargeBOP(is_sub_prefetcher=True)
             prefetcher.bop_small = XSPhysicalSmallBOP(is_sub_prefetcher=True)
         if options.l1_to_l2_pf_hint:
```

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- 3. Declare the prefetcher in SConscript

```
diff --git a/src/mem/cache/prefetch/SConscript b/src/mem/cache/prefetch/SConscript
index c77c5494ad..61756a1bf6 100644
--- a/src/mem/cache/prefetch/SConscript
+++ b/src/mem/cache/prefetch/SConscript
@@ -37,7 +37,7 @@ SimObject('Prefetcher.py', sim_objects=[
     'SignaturePathPrefetcherV2', 'AccessMapPatternMatching', 'AMPMPrefetcher',
     'DeltaCorrelatingPredictionTables', 'DCPTPrefetcher',
     'IrregularStreamBufferPrefetcher', 'SlimAMPMPrefetcher',
-    'WorkerPrefetcher',
+    'WorkerPrefetcher', 'DespacitoStreamPrefetcher',
     'BOPPrefetcher', 'SBOOEPrefetcher', 'STeMSPrefetcher', 'PIFPrefetcher', 'IPCPrefetcher',
     'CompositeWithWorkerPrefetcher', 'L2CompositeWithWorkerPrefetcher'])

@@ -58,6 +58,7 @@ DebugFlag('CDPHotVpns')
 DebugFlag('CDPdebug')
 DebugFlag('CDPdepth')
 DebugFlag('CMCPrefetcher')
+DebugFlag('DespacitoStreamPrefetcher')

 Source('access_map_pattern_matching.cc')
 Source('base.cc')
@@ -86,4 +87,5 @@ Source('worker.cc')
 Source('cmc.cc')
 Source('composite_with_worker.cc')
 Source('l2_composite_with_worker.cc')
+Source('despacito_stream.cc')
```

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- 4. Implement the prefetching
  - Write the core logic of the prefetcher in the source files:
  - src/mem/cache/prefetch/prefetch1.cc
  - src/mem/cache/prefetch/prefetch1.hh

```
src/mem/cache/prefetch/despacito_stream.cc
```

```
src/mem/cache/prefetch/despacito_stream.hh
```

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- 5. Invoke the prefetcher in the L1/L2 prefetching framework

```
diff --git a/src/mem/cache/prefetch/l2_composite_with_worker.cc b/src/mem/cache/prefetch/l2_composite_with_worker.cc
index 31ba47bf01..b7ede79a17 100644
--- a/src/mem/cache/prefetch/l2_composite_with_worker.cc
+++ b/src/mem/cache/prefetch/l2_composite_with_worker.cc
@@ -16,14 +16,17 @@ L2CompositeWithWorkerPrefetcher::L2CompositeWithWorkerPrefetcher(const L2Composi
       largeBOP(p.bop_large),
       smallBOP(p.bop_small),
       cmc(p.cmc),
+      despacitoStream(p.despacito_stream),
       enableBOP(p.enable_bop),
       enableCDP(p.enable_cdp),
-      enableCMC(p.enable_cmc)
+      enableCMC(p.enable_cmc),
+      enableDespacitoStream(p.enable_despacito_stream)
 {
     cdp->pfLRUFilter = &pfLRUFilter;
     largeBOP->filter = &pfLRUFilter;
     smallBOP->filter = &pfLRUFilter;
     cmc->filter = &pfLRUFilter;
+    despacitoStream->filter = &pfLRUFilter;
     cdp->parentRid = p.sys->getRequestorId(this);
 }

@@ -65,6 +68,9 @@ L2CompositeWithWorkerPrefetcher::calculatePrefetch(const PrefetchInfo &pfi, std:
         largeBOP->calculatePrefetch(pfi, addresses, late && pf_source == PrefetchSourceType::HWP_BOP);
         smallBOP->calculatePrefetch(pfi, addresses, late && pf_source == PrefetchSourceType::HWP_BOP);
     }
+    if (enableDespacitoStream) {
+        despacitoStream->calculatePrefetch(pfi, addresses, late && pf_source == PrefetchSourceType::DespacitoStream);
+    }
 }

 void
@@ -121,6 +127,7 @@ L2CompositeWithWorkerPrefetcher::setParentInfo(System *sys, ProbeManager *pm, Ca
     largeBOP->setParentInfo(sys, pm, _cache, blk_size);
     smallBOP->setParentInfo(sys, pm, _cache, blk_size);
     cmc->setParentInfo(sys, pm, _cache, blk_size);
+    despacitoStream->setParentInfo(sys, pm, _cache, blk_size);
     CompositeWithWorkerPrefetcher::setParentInfo(sys, pm, _cache, blk_size);
 }
```

Summary

- **An effective and calibrated simulator for research on** microarchitecture design.
- **Agile development tools** for performance analysis and correctness checking.

https://github.com/OpenXiangShan/GEM5

Thanks!